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Manuscript Title: **Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by $MnWO_4$**

Dear Editor,

We are pleased to submit our manuscript entitled *"Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by $MnWO_4$"* for your consideration.

In this work, we present a comprehensive strategy to

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We believe our work delivers a compelling combination of electrochemical performance optimization and interfacial mechanical insight, contributing a valuable perspective to the field of metal-anode stabilization. We kindly ask that our manuscript be considered for publication in your esteemed journal.

Thank you for your time and consideration.

Sincerely,

Seung-Mo Lee, Prof. Dr.

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Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by $MnWO_4$

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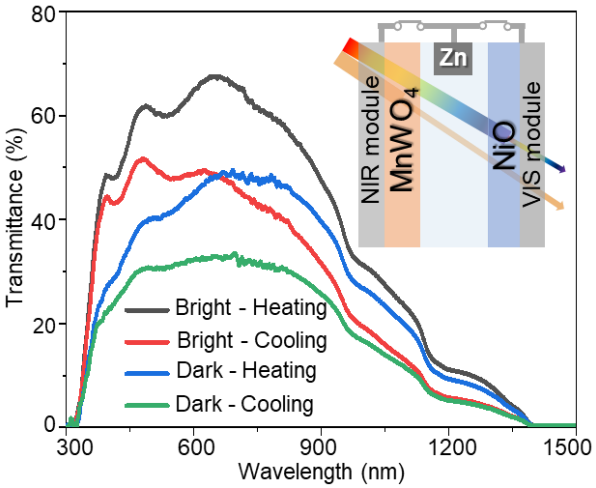
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TOC graphic and summary:

"$MnWO_4$ serves as a multifunctional NIR electrochromic and energy-storage electrode, enabling reversible optical switching and charge storage in a four-mode electrochromic window."



Abstract

Dual-band electrochromic windows that independently regulate visible (VIS) and near-infrared (NIR) solar radiation offer an attractive route to building energy management, yet most reported devices access only three optical states and rarely integrate energy storage. Here we introduce $MnWO_4$ as a dual-function electrode delivering both selective NIR electrochromism and charge storage. In a $MnWO_4$‖Zn‖NiO architecture, a shared Zn anode allows independent addressing of the NIR-active $MnWO_4$ and VIS-active NiO electrodes, unlocking all four operational modes: bright-warm, bright-cool, dark-warm, and dark-cool. The NIR response originates from localized surface plasmon resonance, while storage arises from Zn^{2+} intercalation into the tungstate framework combined with a Mn-centered conversion reaction, giving high areal capacity over 200 cycles. Under simulated sunlight, surface temperatures differ by more than 6 °C between the bright-warm and dark-cool states, establishing an integrated platform for season-adaptive spectral control with on-board energy storage.

Keywords: smart window, electrochromic device, energy storage, $MnWO_4$, four modes

Main Text

Electrochromic (EC) smart windows, which dynamically regulate solar radiation under applied bias, have emerged as a key technology for reducing building energy consumption associated with lighting, heating, and cooling (*1–4*). As buildings account for a substantial fraction of global energy use and carbon emissions, controlling solar heat gain and visible light transmission through windows offers a direct pathway toward improved energy efficiency (*2*, *4*). Solar irradiation primarily consists of visible (VIS, 380–780 nm) and near-infrared (NIR, 780–2500 nm) components, where VIS light governs daylighting and visual comfort, while NIR radiation is responsible for heat transfer (*5*). Consequently, dual-band electrochromic systems capable of independently modulating VIS and NIR transmittance have attracted increasing attention for all-season energy management.

Despite notable progress, most reported dual-band EC devices remain limited in functionality. Conventional systems typically rely on a single electrochromic material or coupled redox processes, leading to synchronized modulation of VIS and NIR regions and restricting operation to three optical states: bright (VIS/NIR transparent), cool (VIS transparent/NIR blocked), and dark (VIS/NIR blocked) (*2*, *6–11*). However, a fully adaptive smart window requires four distinct modes, including an additional “warm” state (VIS blocked/NIR transparent), enabling independent control over lighting and thermal radiation (*12–14*). Achieving such four-mode operation remains challenging due to the difficulty in decoupling VIS and NIR responses while maintaining electrochemical compatibility and device stability (*13*).

Another emerging direction in electrochromic research is the integration of energy storage functionality into EC platforms (*15–17*). Electrochromic energy-storage devices, including electrochromic batteries and supercapacitors, offer the dual advantage of modulating optical properties while storing and delivering electrical energy. This integration is particularly attractive for self-powered smart windows, where harvested or externally supplied energy can be stored and reused, thereby enhancing overall system efficiency. However, the combination

of materials and device architectures that simultaneously enable dual-band optical control and efficient energy storage remains well developed.

From a materials perspective, transition metal oxides have dominated EC research, particularly for NIR modulation via localized surface plasmon resonance or polaronic absorption (*18*). In contrast, transition metal tungstates remain relatively unexplored despite their rich redox chemistry and structural diversity. Among them, $MnWO_4$ is a promising candidate due to the coexistence of Mn redox centers and a tungsten-based framework, which can facilitate ion intercalation, electron transport, and charge compensation (*19*, *20*). The multivalent nature of manganese (Mn^{2+}/Mn^{3+}) enables reversible Faradaic reactions that are not only beneficial for electrochromic modulation, especially in the NIR region, but also contribute to pseudocapacitive energy storage. This dual functionality makes $MnWO_4$ particularly attractive for coupling electrochromism with energy storage within a single material platform. Meanwhile, NiO is a well-established anodic electrochromic material exhibiting stable and reversible coloration in the VIS region, making it an ideal counterpart for selective visible-light modulation.

Herein, we report a dual-band electrochromic energy-storage device that integrates $MnWO_4$ and NiO electrodes to achieve independent regulation of NIR and VIS light. By combining $MnWO_4$ as the NIR-active electrode and NiO as the VIS-active electrode within a unified electrochemical system, the device enables four distinct optical modes: bright (VIS/NIR transparent), cool (NIR blocked/VIS transparent), warm (VIS blocked/NIR transparent), and dark (VIS/NIR blocked). Importantly, the redox processes governing electrochromism also contribute to charge storage, allowing the device to function as an energy-storage unit. These findings establish $MnWO_4$ as a promising multifunctional electrode material and highlight its potential for integration into next-generation multifunctional smart window systems that combine energy storage, electrochromic regulation, and enhanced building energy efficiency.

Design Principles of the Four-Mode Electrochromic Device

The design concept of the four-mode electrochromic window is illustrated in Fig. 1. The device adopts a three-electrode architecture in which a $MnWO_4$ NIR module and a NiO VIS module are independently addressable through a shared Zn anode. This configuration enables decoupled electrochemical control of the two complementary electrodes, which is the key prerequisite for achieving four distinct operational modes. In the bright–warm mode, both electrodes are in the bleached state, allowing high transmittance of both VIS and NIR radiation simultaneously (Fig. 1a). This mode is suitable for cold seasons or low-light conditions where maximum solar heat gain and indoor illumination are desirable. In the bright–cool mode, the NiO VIS module is colored while $MnWO_4$ remains bleached, selectively blocking visible light while permitting NIR transmission (Fig. 1b). In the dark–warm mode, $MnWO_4$ is colored to block NIR radiation while NiO remains bleached, preserving daylight transmission while reducing solar heat gain (Fig. 1c). This mode is particularly valuable in hot seasons where thermal control is needed without sacrificing natural lighting. Finally, in the dark–cool mode, both electrodes are colored, resulting in simultaneous blocking of VIS and NIR radiation for maximum solar rejection (Fig. 1d). The realization of all four modes, particularly the elusive dark–warm state, distinguishes this device from most reported dual-band electrochromic windows, which are typically limited to three optical states. The independent control of VIS and NIR transmittance enabled by the three-electrode Zn-anode architecture is thus central to the versatility and season-adaptive capability of the device.

Pristine WO_3 adopts a classical three-dimensional framework of corner-sharing WO_6 octahedra (Fig. 1e). While this sustains reliable electrochromic modulation, the specific capacitance remains inherently low: charge storage in standalone WO_3 is confined by bulk ion diffusion and restricted to the available crystallographic voids, preventing it from serving as an active pseudocapacitive matrix (*21*). $MnWO_4$, in contrast, adopts a monoclinic wolframite chain configuration of edge-sharing MnO_6 and WO_6 octahedra (Fig. 1f). This open arrangement exposes abundant surface active sites to the electrolyte, and the multiple Mn oxidation states

enable fast, reversible Faradaic redox at the surface and near-surface region. We believe that $MnWO_4$ combines excellent electrochromism with robust pseudocapacitive storage, marking it as a true bifunctional smart-window electrode.

Characterization of MnWO4

$MnWO_4$ was synthesized using a facile hydrothermal method (Fig. S1). Its structural, morphological, and chemical properties were systematically characterized. Scanning electron microscopy (SEM) imaging revealed a nanowire-like morphology with densely interconnected features distributed uniformly across the film surface (Fig. 2a). This porous, high-surface-area microstructure is expected to facilitate electrolyte penetration and ion transport during both electrochromic switching and energy storage operation. Low-magnification transmission electron microscopy (TEM) analysis further confirmed the nanowire-like morphology (Fig. 2b). The high-resolution TEM image exhibited clear lattice fringes with an interplanar spacing of 3.75 Å, corresponding to the (110) crystal plane of $MnWO_4$ (Fig. 2c) (*22*). This assignment was further corroborated by the selected area electron diffraction (SAED) pattern (Fig. S2). The X-ray diffraction (XRD) pattern confirmed the phase purity of the as-synthesized $MnWO_4$, with all diffraction peaks in good agreement with the monoclinic wolframite crystal structure (Fig. 2d and Fig.S3). No secondary phases were detected, indicating a clean and well-controlled synthesis. X-ray photoelectron spectroscopy (XPS) was performed to examine the valence states of tungsten and manganese. The high-resolution W 4f spectrum exhibited two sets of peaks assigned to W^{6+} $4f_{5/2}$ and W^{6+} $4f_{7/2}$, confirming that tungsten is predominantly in the +6 oxidation state (Fig. 2e). The Mn 2p spectrum showed characteristic peaks corresponding to Mn^{2+} $2p_{1/2}$ and Mn^{2+} $2p_{3/2}$, accompanied by satellite features consistent with high-spin divalent manganese in an octahedral coordination environment (Fig. 2f) (*23*). These results collectively confirm the successful formation of stoichiometric, phase-pure $MnWO_4$ with the expected electronic structure, providing a well-defined starting point for investigating its electrochromic and energy storage properties.

Energy Storage Performance of MnWO4

The electrochemical energy storage behavior of $MnWO_4$ was evaluated in a Zn‖$MnWO_4$ cell. The CV curve recorded at a scan rate of 0.1 mV s^{-1} revealed multiple distinct redox features, indicating a complex, multi-step electrochemical process (Fig. 3a). Two cathodic reactions were identified upon discharge. At moderate potentials, Zn^{2+} ions intercalate into the WO_6 octahedra of $MnWO_4$ to form Zn_xWO_3, contributing to capacity through an intercalation mechanism. At lower potentials, a conversion reaction occurs in which MnO_6 octahedra is reduced by Zn^{2+} to form Zn_xMnO. The corresponding anodic process upon charging involves the decomposition of Zn_xMnO back to MnO, confirming the reversibility of the conversion reaction. The coexistence of intercalation and conversion pathways within a single electrode material is notable, as it enables $MnWO_4$ to deliver higher capacity than materials relying on a single storage mechanism alone. In contrast, the CV curve of commercial WO_3 exhibited only one broad peak (Fig. S4), indicating limited redox chemistry of WO_3. Galvanostatic charge-discharge profiles measured at current densities ranging from 0.01 to 0.1 mA cm^{-2} demonstrated well-defined plateaus consistent with the redox processes identified in the CV (Fig. 3b). The cell delivered an areal capacity of approximately 2 mAh cm^{-2} at the lowest current density of 0.01 mA cm^{-2}, reflecting the high electrochemical utilization of the $MnWO_4$ electrode. Rate capability testing showed a gradual decrease in capacity with increasing current density, retaining a meaningful capacity at 0.1 mA cm^{-2}, which is indicative of reasonably fast ion transport kinetics (Fig. 3c). Long-term cycling stability was assessed at both 0.02 and 0.1 mA cm^{-2} over 200 consecutive cycles (Fig. 3d–e). At both current densities, the areal capacity remained stable throughout cycling with coulombic efficiency maintained close to 100%, demonstrating that the $MnWO_4$ electrode undergoes highly reversible electrochemical reactions without significant capacity fade. These results establish $MnWO_4$ as a capable energy storage electrode material, complementing its electrochromic functionality within the integrated device.

Electrochromic Performance of $MnWO_4$

The electrochromic properties of $MnWO_4$ were characterized in the Zn‖$MnWO_4$ cell configuration. Digital photographs captured at different applied voltages clearly demonstrated a reversible and pronounced color transition (Fig. 4a). At 1.2 V, the electrode appeared in a light, near-transparent bleached state. Upon reducing the voltage to 0.0 V, the electrode darkened significantly into a deep blue-grey colored state. The original bleached appearance was fully recovered upon reapplying 1.2 V, confirming excellent optical reversibility. Transmittance spectra measured across the 300–1500 nm range at 0.0, 0.5, and 1.2 V revealed that the coloration effect was most pronounced in the NIR spectral region, with a large transmittance contrast between the colored and bleached states (Fig. 4b). The VIS transmittance, while also modulated, changed to a lesser extent relative to the NIR range. This spectral selectivity toward NIR modulation is consistent with an LSPR-driven electrochromic mechanism, in which free carrier absorption at low potentials preferentially attenuates NIR radiation before the onset of deeper polaronic coloration in the VIS range. The intermediate transmittance profile at 0.5 V further illustrates the gradual and voltage-tunable nature of the NIR modulation. The long-term optical stability of the $MnWO_4$ electrode was assessed by comparing transmittance spectra before and after 500 coloration-bleaching cycles (Fig. 4c). The spectra at both the colored and bleached states after 500 cycles closely overlapped with those recorded at cycle 1, demonstrating negligible degradation in optical contrast and spectral character. This electrochromic durability, combined with the energy storage stability demonstrated in Fig. 3, underscores the robustness of $MnWO_4$ as a dual-function electrode material.

Four-Mode Electrochromic Device Performance

The complete $MnWO_4$‖Zn‖NiO three-electrode device was assembled, and its four-mode optical operation was systematically characterized. The $MnWO_4$ electrode is responsible for

NIR regulation, while NiO electrode serves as VIS module (Fig. S5–S7). It should be noted that NiO exhibits limited stability. Therefore, improving its cycling stability should be a key focus of future work. Digital photographs of the device in each of the four modes showed clearly distinguishable optical appearances, confirming that each mode produces a visually distinct state (Fig. 5a). The bright–warm mode appeared the lightest and most transparent, while the dark–cool mode was the darkest. The bright–cool and dark–warm modes exhibited intermediate and spectrally differentiated appearances, reflecting the selective coloration of either the NiO VIS module or the $MnWO_4$ NIR module, respectively. Broadband transmittance spectra across the 300–1500 nm range provided quantitative confirmation of the spectral selectivity achieved in each mode (Fig. 5b). The bright–warm state exhibited the highest transmittance across both the VIS and NIR bands. In the bright–cool state, transmittance in the VIS region was substantially reduced while NIR transmittance remained comparatively high, consistent with selective NiO coloration. In the dark–warm state, NIR transmittance was preferentially attenuated by MnWO4 coloration while VIS transmittance was better preserved, demonstrating the sought-after spectral inversion relative to the bright–cool mode. The dark–cool state showed the lowest transmittance across both bands, corresponding to simultaneous coloration of both electrodes. The ability to independently switch between these four spectrally distinct states within a single device represents a significant advance over conventional three-mode dual-band EC windows. The practical thermal management utility of the device was validated through solar simulator experiments monitored by infrared thermal imaging (Fig. 5c). A black heat-absorbing textile placed beneath the device was illuminated for up to 600 s. For comparison, under normal glass, the surface temperature increased rapidly, reaching 42.1 °C after 600 s of illumination (Fig. S8). In the bright–warm mode, the textile temperature rose from an initial 24.5 °C to 32.1 °C after 10 s, 34.4 °C after 60 s, and 37.2 °C after 600 s of continuous illumination (Fig. 5d). In contrast, the dark–cool mode markedly suppressed heat accumulation under identical conditions, with the temperature reaching only 29.2 °C after 10 s, 30.7 °C after 60 s, and 31.1 °C after 600 s. Notably, after 600 s, the temperature beneath

the electrochromic device in the dark–cool mode was approximately 11 °C lower than that under normal glass and 6 °C lower than that in the bright–warm mode. These results clearly demonstrate the device's effectiveness in regulating solar heat gain and reducing thermal load. These results highlight the potential of the MnWO4‖Zn‖NiO device as a practical, multifunctional smart window that combines season-adaptive solar regulation with integrated electrochemical energy storage.

Conclusion

In summary, we demonstrated a four-mode dual-band electrochromic window with integrated energy storage based on a $MnWO_4$‖Zn‖NiO three-electrode architecture. $MnWO_4$ was introduced for the first time as a dual-function electrode material, delivering selective NIR electrochromic modulation through an LSPR-driven mechanism alongside high areal capacity through combined Zn^{2+} intercalation and conversion reactions. The independent electrochemical addressability of the $MnWO_4$ NIR module and NiO VIS module enabled full four-mode operation, bright–warm, bright–cool, dark–warm, and dark–cool, including the elusive dark–warm state that remains inaccessible in most reported dual-band devices. Infrared thermal imaging confirmed a surface temperature difference of over 6 °C between the bright–warm and dark–cool modes under simulated solar illumination, validating the practical thermal regulation capability of the device. The integration of season-adaptive spectral control and on-board energy storage within a single platform represents a meaningful step toward self-sustaining, energy-efficient building envelope technologies.

Acknowledgments

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Conflict of Interest

The authors declare no conflicts of interest.

Supporting Information

The Supporting Information is available at …

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Figures

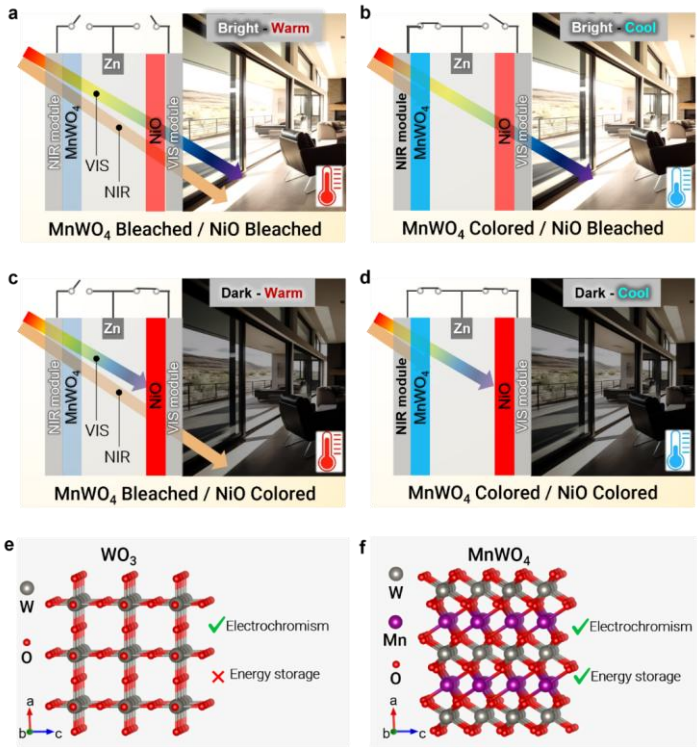


Figure 1. Operating principle of the four-mode electrochromic window, and why $MnWO_4$ can store charge while WO_3 cannot. (a–d) Schematic of the three-electrode device. An NIR-active $MnWO_4$ electrode and a VIS-active NiO electrode share one Zn electrode via independent switches, so each can be switched on (colored) or off (bleached) separately. Arrows trace the VIS and NIR components of transmitted sunlight, and the room scenes illustrate the resulting effect. **(a)** Bright–warm (both electrodes bleached). VIS and NIR are both transmitted for cold or low-light conditions. **(b)** Bright–cool (NiO colored, $MnWO_4$ bleached). VIS is blocked, and NIR (heat) is still admitted. **(c)** Dark–warm ($MnWO_4$ colored, NiO bleached). NIR (heat) is blocked, and VIS is admitted for hot sunny days without darkening the room. **(d)** Dark–cool (both colored). VIS and NIR are both blocked for maximum solar rejection. **(e)** Crystal structures of WO_3, built from corner-sharing WO_6 octahedra in a dense, fully connected framework. **(f)** $MnWO_4$, built from more open, edge-sharing MnO_6/WO_6 chains. WO_3's framework supports electrochromism but leaves little room for ion storage. In contrast, $MnWO_4$'s exposed Mn sites support both electrochromism and genuine charge storage, as summarized by the checkmarks.

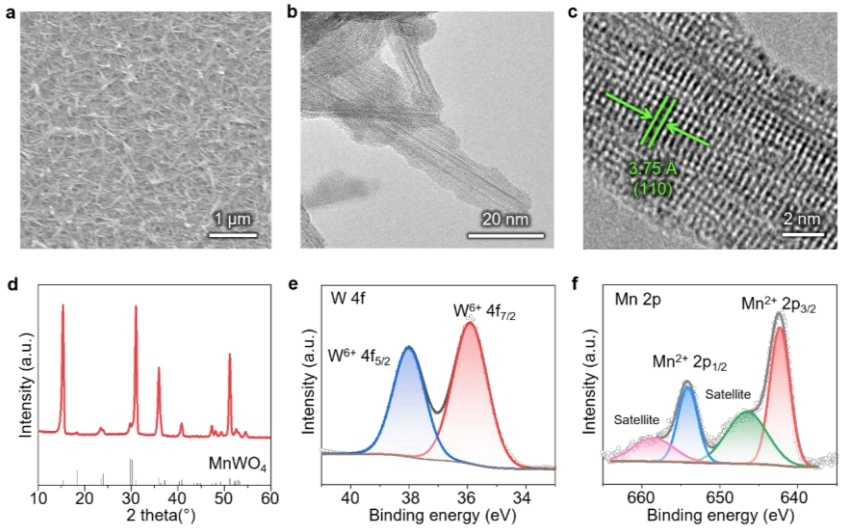


Figure 2. Structural and chemical confirmation of the as-synthesized $MnWO_4$ nanowire film. **(a)** SEM image showing a dense, uniform mat of $MnWO_4$ nanowires on ITO. A porous morphology is expected to aid electrolyte penetration and ion transport. **(b)** Low-magnification TEM image confirming the nanowire morphology at the single-particle level. **(c)** High-resolution TEM image showing continuous lattice fringes spaced 3.75 Å apart, matching the (110) plane of monoclinic $MnWO_4$ (22) and confirming the nanowires are single-crystalline. (d) XRD pattern (red) matched against the reference pattern for monoclinic wolframite $MnWO_4$ (grey bars). Every peak is accounted for with no impurity phases. (e) High-resolution XPS of the W 4f region. The spin-orbit-split $4f_{5/2}$, $4f_{7/2}$ doublet confirms tungsten is entirely in the W^{6+} state before any coloring reaction. (f) High-resolution XPS of the Mn 2p region. The Mn^{2+} $2p_{3/2}$, $2p_{1/2}$ peaks and their satellite features confirm high-spin Mn^{2+} in an octahedral oxygen site, consistent with the structure in Fig. 1f (23).

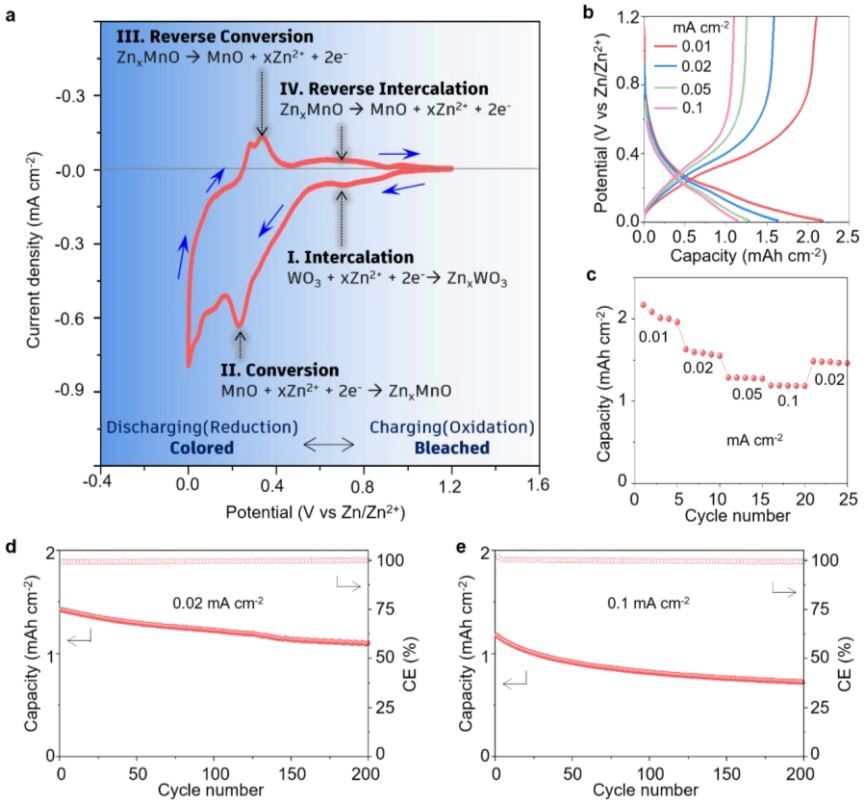


Figure 3(revision needed). Energy-storage performance of $MnWO_4$, tested as a battery-type electrode against a Zn counter-electrode. **(a)** Cyclic voltammogram of the Zn‖$MnWO_4$ cell (0.1 mV s^{-1}). On discharge, Zn^{2+} first intercalates into the WO_6 framework to form Zn_xWO_3, then at lower potential converts the MnO_6 framework to Zn_xMnO. Both reactions reverse on charging, as labeled directly on the curve. **(b)** Galvanostatic charge–discharge curves at 0.01–0.1 mA cm^{-2}, with plateaus matching the reactions in **(a)**. The slowest rate delivers the largest areal capacity, ≈2 mAh cm^{-2}. **(c)** Rate capability over 25 cycles as the current density is stepped up and back down. This shows that capacity is largely recovered after fast cycling. **(d,e)** Cycling stability over 200 cycles at (d) 0.02 and (e) 0.1 mA cm^{-2}, showing stable areal capacity (left axis) and coulombic efficiency near 100% (right axis) throughout.

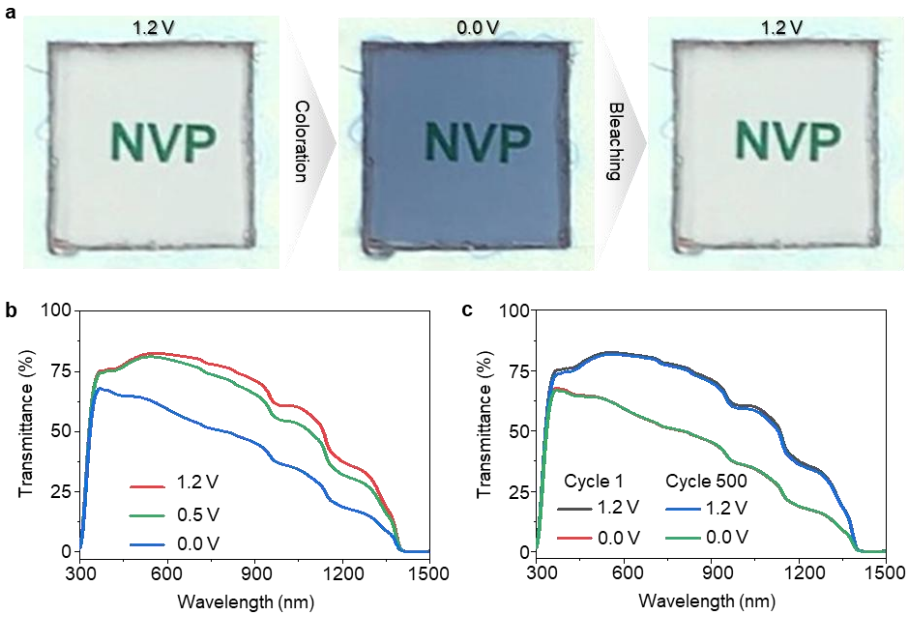


Figure 4. Electrochromic (color-switching) performance of the $MnWO_4$ electrode in the same Zn‖$MnWO_4$ cell. (a) Photographs (green "NVP" logo as a transparency reference) at 1.2 V (bleached, logo legible), 0.0 V (colored, deep blue-grey, logo obscured), and back at 1.2 V (bleached state fully restored), demonstrating reversible switching. **(b)** Transmittance spectra (300–1500 nm) at 1.2, 0.5, and 0.0 V. The colored–bleached contrast is far larger in the near-infrared (>780 nm) than in the visible range, showing that $MnWO_4$ selectively blocks heat-carrying NIR light, consistent with a plasmon-like (LSPR) mechanism. The 0.5 V curve shows that the modulation is continuously voltage-tunable. **(c)** Transmittance spectra at the colored and bleached states, compared between cycle 1 and cycle 500. The curves overlap closely, showing negligible loss of optical contrast after 500 cycles.

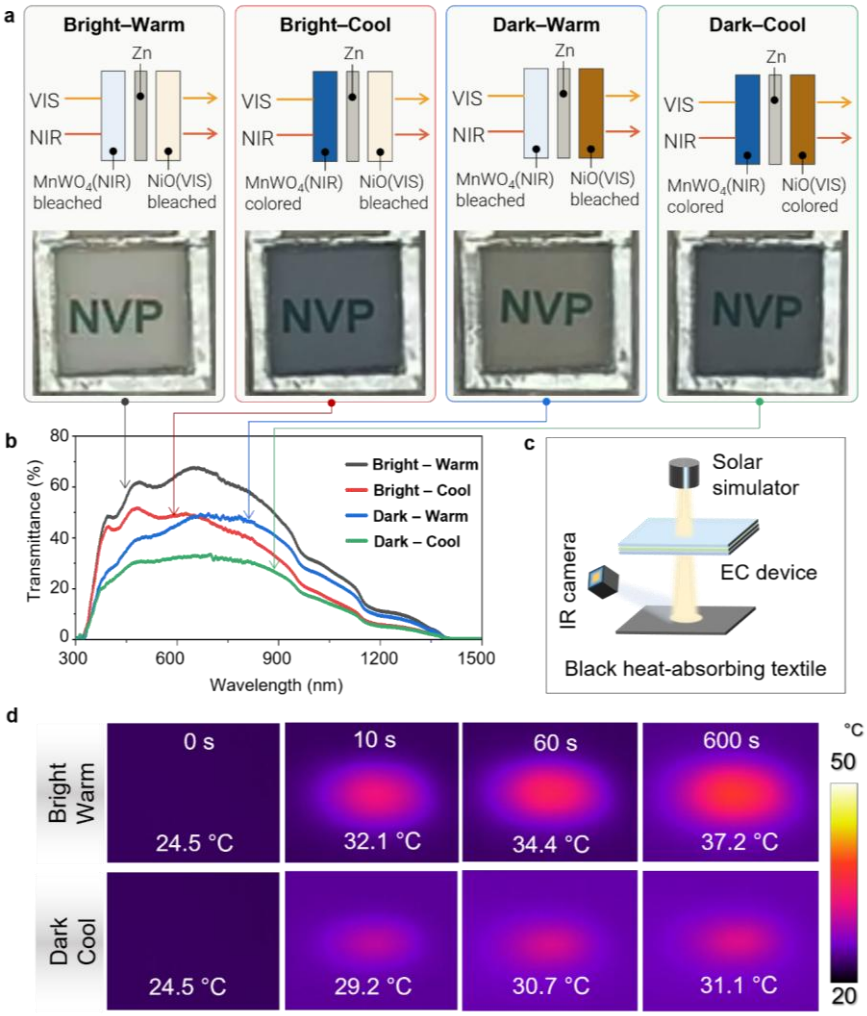


Figure 5. Performance of the complete four-mode $MnWO_4$‖Zn‖NiO device, including a solar heat-blocking demonstration. **(a)** Photographs of the assembled device in each of the four modes (same logo reference as Fig. 4a). Bright–warm is lightest/most transparent. Dark–cool is darkest, and bright–cool/dark–warm show visually distinct intermediate colorations from NiO and $MnWO_4$, respectively. **(b)** Transmittance spectra (300–1500 nm) confirming this quantitatively. Bright–warm gives the highest transmittance in both bands; bright–cool suppresses VIS while preserving NIR; dark–warm shows the mirror-image selectivity; Dark–cool suppresses both. **(c)** Test setup. A solar simulator illuminates the device, a black heat-

absorbing textile behind it mimics an interior surface, and an infrared camera tracks its temperature. **(d)** Infrared thermal images of the textile in bright–warm (top) and dark–cool (bottom) modes at 0, 10, 60, and 600 s. Bright–warm heats steadily (24.5→37.2 °C); dark–cool suppresses warming (24.5→31.1 °C) after 600 s and is ~6 °C cooler than bright–warm and ~11 °C cooler than ordinary glass (Fig. S8), showing the modes make a practically meaningful difference in transmitted heat.

SUPPORTING INFORMATION

Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by $MnWO_4$

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1. Experimental Section

메모 포함[SML2]: Phuong, if possible, we had better describe the experimental procedure in detail.

1.1. Materials

All chemicals were used as received without further purification unless noted otherwise. Sodium tungstate dihydrate ($Na_2WO_4 \cdot 2H_2O$), manganese(II) chloride tetrahydrate ($MnCl_2 \cdot 4H_2O$), sodium sulfate (Na_2SO_4), hydrochloric acid (HCl), and zinc sulfate ($ZnSO_4$) were used for $MnWO_4$ synthesis and electrolyte preparation. ITO-coated glass substrates served as the transparent conducting substrate for both electrodes. Zinc foil/mesh was used as the shared counter-electrode.

메모 포함[SML3]: $Mn(CH_3COO)_2 \cdot 4H_2O$ is correct or $(MnCl_2 \cdot 4H_2O$ is correct?

메모 포함[SML4]: If possible, please insert supplier information. Or company.

메모 포함[SML5]: If possible, please insert supplier information. Or company.

1.2. Synthesis of MnWO4 nanowire film on ITO glass

$MnWO_4$ was synthesized directly onto ITO glass by a one-step hydrothermal method. Briefly, 0.3 mmol $Na_2WO_4 \cdot 2H_2O$, 0.3 mmol $MnCl_2 \cdot 4H_2O$, and 0.1 mmol Na_2SO_4 (a 1:1 Mn : W molar ratio, with Na_2SO_4 added as a mineralizing/morphology-directing additive) were dissolved in 40 mL deionized water under magnetic stirring. The solution pH was adjusted to ≈2 by the dropwise addition of HCl, and the mixture was stirred for 1 h at room temperature to ensure complete homogenization. The precursor solution was then transferred into a 50 mL Teflon (PTFE)-lined stainless-steel autoclave. ITO glass substrates, pre-cleaned by sequential ultrasonication in ethanol and deionized water and dried under a nitrogen/air stream, were immersed vertically in the precursor solution inside the autoclave. The sealed autoclave was heated at 180 °C for 1.5 h in a temperature-controlled convection oven, then allowed to cool naturally to room temperature. The $MnWO_4$-coated ITO substrates were removed, rinsed

thoroughly with deionized water and ethanol to remove residual salts, and dried under ambient conditions.

1.3 Preparation of NiO-coated ITO glass

The NiO on ITO glass was prepared by sputtering method. …

1.4 Material characterizations

The surface morphology and elemental composition of the samples were characterized using field-emission scanning electron microscopy (FE-SEM, JSM-7800F, JEOL) equipped with an energy-dispersive X-ray spectroscopy (EDS) system (AZtec®, Oxford Instruments). Crystal structures were examined by X-ray diffraction (XRD) using an Empyrean diffractometer (PANalytical) with Cu Kα radiation (λ = 1.5418 Å). The chemical composition and surface electronic states were investigated by X-ray photoelectron spectroscopy (XPS) using a MultiLab 2000 system (Thermo Fisher Scientific) equipped with an Al Kα X-ray source. The microstructure of the $MnWO_4$ samples was further analyzed by transmission electron microscopy (TEM, Talos G2, Thermo Fisher Scientific) operated at an accelerating voltage of 200 kV.

1.5 Device assembly (half-cells and full three-electrode device)

Both single-electrode (half-cell) test cells and the complete three-electrode $MnWO_4$‖Zn‖NiO device were assembled in a sandwich configuration. For the half-cells (Zn‖$MnWO_4$ or Zn‖NiO), a 1-mm-thick double-sided adhesive tape (3M) with a rectangular window (1.5 × 1.5 cm^2 active area) cut into its center was laminated onto the $MnWO_4$- or NiO-coated ITO substrate, so that the tape simultaneously defined the active-area spacer and provided an edge seal. A zinc foil electrode pre-mounted on a plain glass cover slide was then aligned over the tape to close the cell, and the resulting cavity was filled with 0.5 M aqueous $ZnSO_4$ electrolyte, injected through a small unsealed gap with a syringe. The complete three-electrode device was assembled using the same lamination procedure, but with a zinc frame electrode sandwiched between one $MnWO_4$-coated ITO substrate and one NiO-coated ITO substrate, each contacted independently so that the two electrodes could be addressed separately against the shared Zn electrode, giving the three-terminal wiring shown schematically in Fig. 1a–d.

메모 포함[SML6]: Phuong, we had better clarify which one you used, foil or mesh?

1.6 Electrochemical and electrochromic testing

All electrochemical measurements were performed on a VSP potentiostat/galvanostat (Bio-Logic Science Instruments) at room temperature in ambient air, using a two-electrode configuration in which the $MnWO_4$- or NiO-coated ITO substrate served as the working electrode and the Zn foil served as the combined counter and reference electrode, so that all reported potentials are versus Zn/Zn^{2+}. Cyclic voltammetry was recorded at a scan rate of 0.1 mV s^{-1} (Fig. 3a, Fig. S4). Galvanostatic charge–discharge and rate-capability tests were carried out at current densities of 0.01, 0.02, 0.05, and 0.1 mA cm^{-2} (Fig. 3b,c). Long-term cycling of the energy-storage performance was carried out for 200 consecutive charge–discharge cycles at 0.02 mA cm^{-2} (Fig. 3d) and, separately, at 0.1 mA cm^{-2} (Fig. 3e). Electrochromic (coloration/bleaching) cycling durability was assessed by switching the $MnWO_4$ half-cell between its bleached (1.2 V) and colored (0.0 V) states for 500 cycles (Fig. 4c), and the NiO half-cell between its bleached (1.0 V) and colored (2.3 V) states for 30 cycles (Fig. S7). Areal capacity was calculated as Q (mAh cm^{-2}) = I × t / A, where I is the applied current (mA), t is the discharge time (h), and A is the geometric active-device area (1.5 × 1.5 cm^2). Coulombic efficiency was calculated as the discharge areal capacity divided by the charge areal capacity, expressed as a percentage. Each of the four optical modes of the complete device (Fig. 5a,b) was set by independently biasing the $MnWO_4$ electrode to its bleached (1.2 V) or colored (0.0 V) state and the NiO electrode to its bleached (1.0 V) or colored (2.3 V) state, using the shared Zn electrode as the common reference for both.

메모 포함[SML7]: Phuong, we had better clarify which one you used, foil or mesh?

메모 포함[SML8]: I describe based on my guess. If something is wrong, please make a correction.

1.7 Solar-simulator thermal imaging

A solar simulator was used to illuminate the assembled device from above, with a black heat-absorbing textile placed directly behind the device to mimic an interior surface (Fig. 5c). The surface temperature of the textile was recorded with an infrared (thermal) camera at 0, 10, 60, and 600 s after the start of illumination [write here…what solar simulator?, AM 1.5G, 1-sun = 100 mW cm^{-2} ….conditions, temperature etc..]. The same protocol, without the electrochromic device present (bare glass only), was used to obtain the reference curve in Fig. S8.

메모 포함[SML9]: Please fill in..

2. Supporting Figures

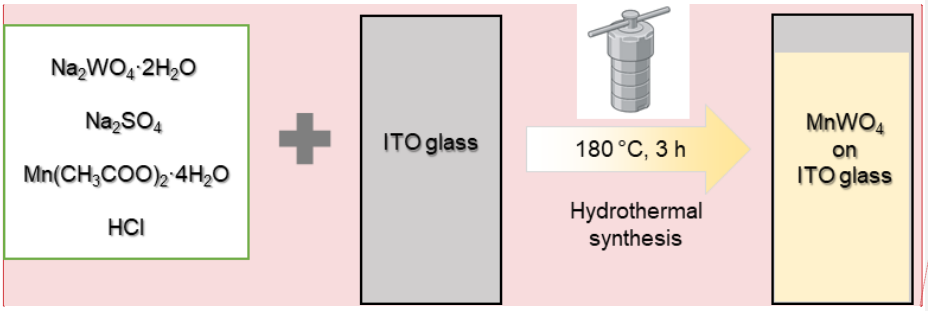


Figure S1. Hydrothermal synthesis route for $MnWO_4$ films on ITO glass. Schematic flow diagram of the one-step synthesis. The precursor reagents ($Na_2WO_4 \cdot 2H_2O$, Na_2SO_4, and the manganese precursor ($Mn(CH_3COO)_2 \cdot 4H_2O$), together with HCl for pH adjustment) are combined with an ITO glass substrate inside a sealed autoclave and heated at 180 °C. After the hydrothermal step, the substrate emerges coated with a uniform $MnWO_4$ nanowire film (pale, translucent coating), ready for rinsing and drying.

메모 포함[SML10]: Among the chemicals, Mn(CH3COO)2 4H2O is correct or ($MnCl_2 \cdot 4H_2O$) is correct?

In the experimental, you mentioned that you used $MnCl_2 \cdot 4H_2O$

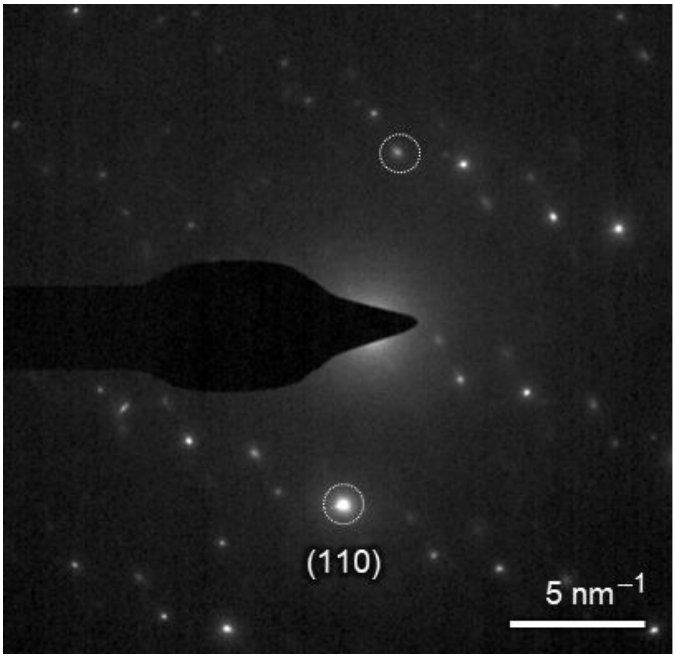
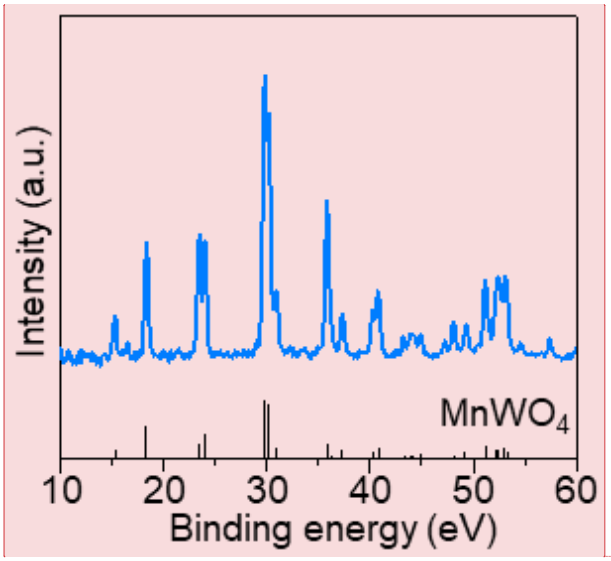


Figure S2. Selected-area electron diffraction (SAED) pattern of a $MnWO_4$ nanowire. Electron diffraction pattern collected on an individual nanowire (dark triangular shadow at image center is the sample holder/beam stop). The bright diffraction spots form a regular pattern consistent with a single-crystalline (rather than amorphous or randomly polycrystalline) nanowire. The spot circled and labeled (110) corresponds to the same lattice plane measured directly by lattice-fringe spacing in Fig. 2c (3.75 Å), cross-confirming that assignment by an independent method.



메모 포함[SML11]: The x-label must be "2θ (°)" not binding energy.

Figure S3. XRD pattern of bulk $MnWO_4$ powder collected from the autoclave liner wall. XRD pattern (blue trace) of $MnWO_4$ powder that nucleated on the inner wall of the PTFE autoclave liner during the same hydrothermal synthesis batch used for the ITO thin films. Vertical reference bars mark the standard peak positions for monoclinic $MnWO_4$. All measured peaks align with the reference pattern, and no unassigned peaks are present, corroborating the thin-film phase assignment in Fig. 2d.

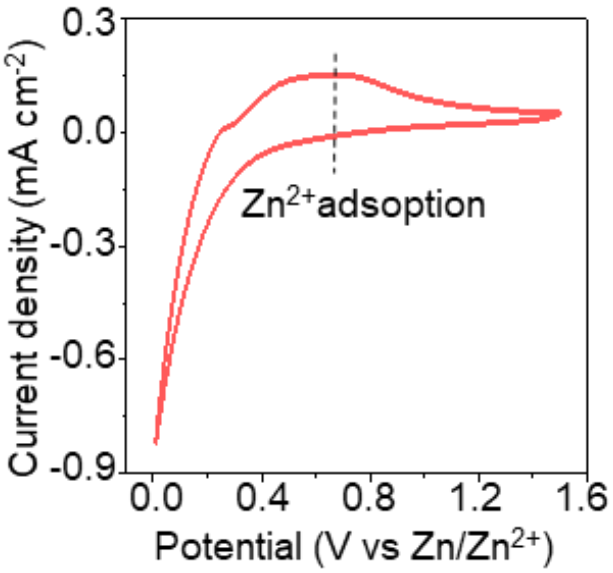


Figure S4. Cyclic voltammogram of commercial WO_3, tested for comparison against $MnWO_4$. CV curve of a commercial WO_3 electrode (Zn‖WO_3 cell, 0.1 mV s^{-1}, same conditions as Fig. 3a) showing a single, broad redox feature associated with Zn^{2+} adsorption/insertion, in clear contrast to the multiple distinct, sharper redox features seen for $MnWO_4$ in Fig. 3a. This comparison supports the claim in the main text that $MnWO_4$ has substantially richer, more multi-step redox chemistry than conventional WO_3, underlying its higher energy-storage capacity.

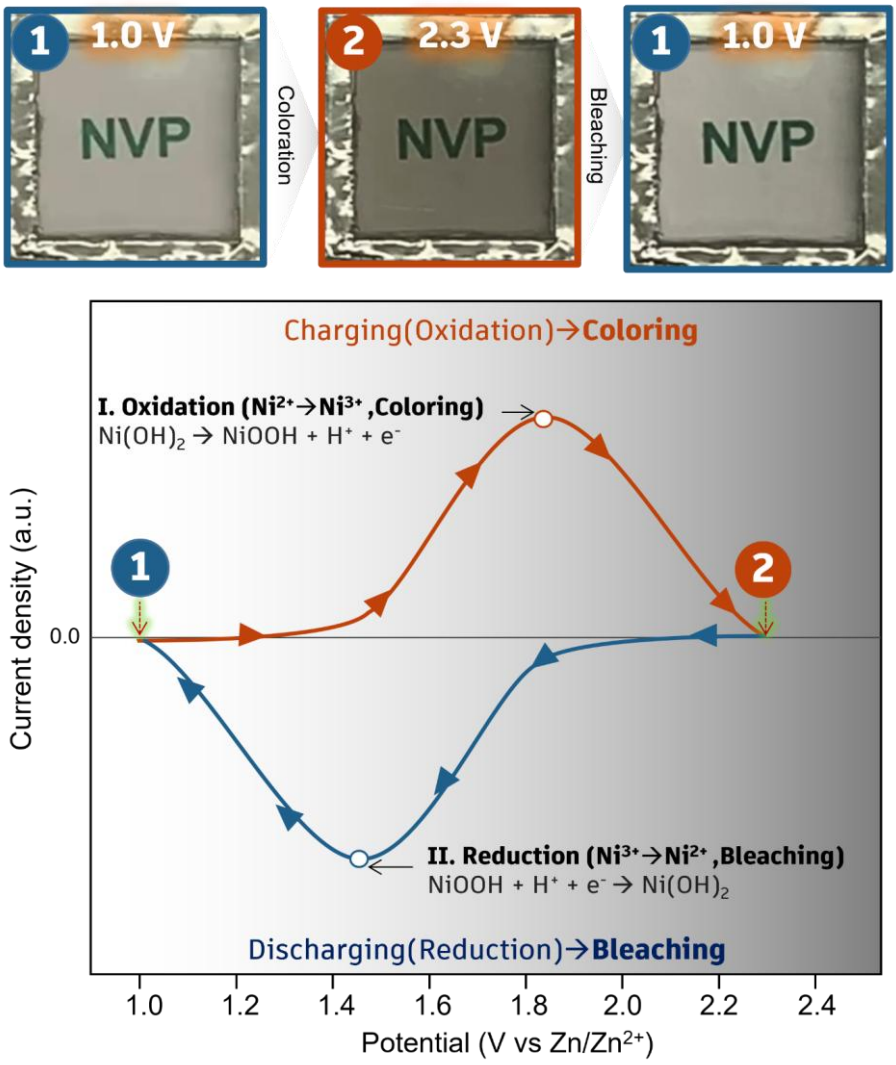
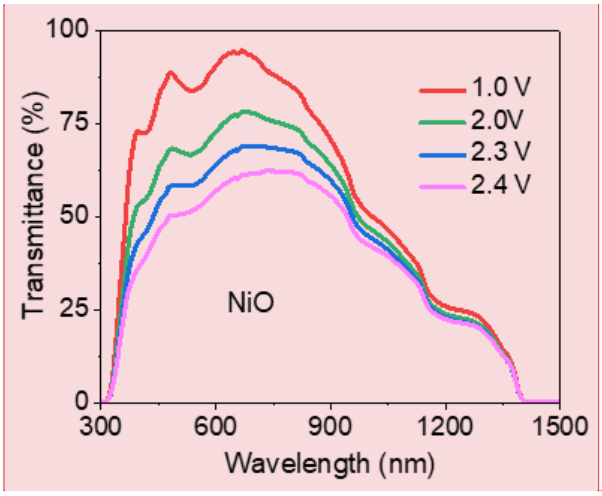


Figure S5(revision needed). Reversible electrochromic switching of the NiO half-cell. Digital photographs of the Zn||NiO cell (same "NVP" logo transparency reference as Fig. 4a) at 1.0 V (bleached, light tan/grey, logo clearly legible), after switching to 2.3 V (colored, dark brownish-grey, logo much harder to read), and after returning to 1.0 V (bleached state restored), confirming that, like $MnWO_4$, the NiO electrode's coloration is optically reversible.



메모 포함[SML12]: 2.0V → 2.0 V

Figure S6. Voltage-dependent transmittance of the NiO half-cell. Transmittance spectra (300–1500 nm) of the Zn‖NiO cell at four voltages (1.0, 2.0, 2.3, and 2.4 V, color-coded). Transmittance decreases progressively across the visible-light-dominated portion of the spectrum as voltage increases from 1.0 to 2.4 V, confirming that NiO coloration is primarily a visible-light-modulating (rather than near-infrared-selective) effect — the complementary spectral behavior to $MnWO_4$ (Fig. 4b), which is what allows the two electrodes to be combined for independent VIS/NIR control in the full four-mode device.

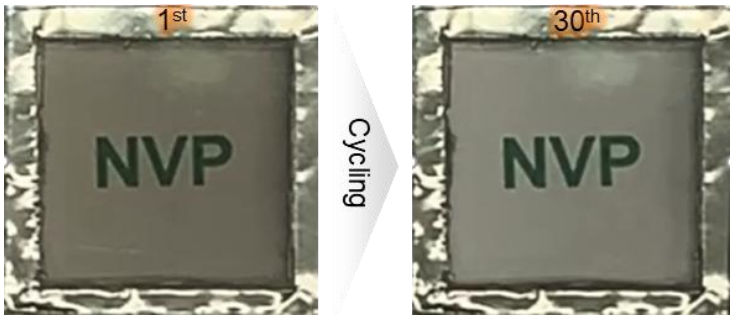


Figure S7. Cycling stability of the NiO half-cell. Digital photographs comparing the colored-state appearance of the Zn‖NiO cell at the 1st and 30th coloration/bleaching cycle. The visibly similar appearance between the two indicates reasonable short-term reversibility.

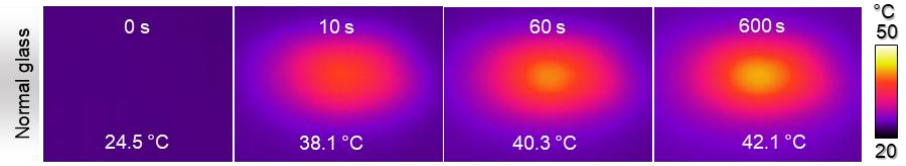


Figure S8. Reference thermal-imaging test with ordinary window glass (no electrochromic device). Infrared thermal images of the black heat-absorbing textile (same setup as Fig. 5c,d) placed behind a sheet of ordinary, uncoated window glass instead of the electrochromic device, at 0, 10, 60, and 600 s of solar-simulator illumination (same false-color temperature scale, 20–50 °C). The textile heats steadily and reaches the highest temperature of any condition tested in this study (24.5 → 38.1 → 40.3 → 42.1 °C), serving as the "no control" baseline against which the bright–warm and dark–cool performance of the actual device (Fig. 5d) is compared.